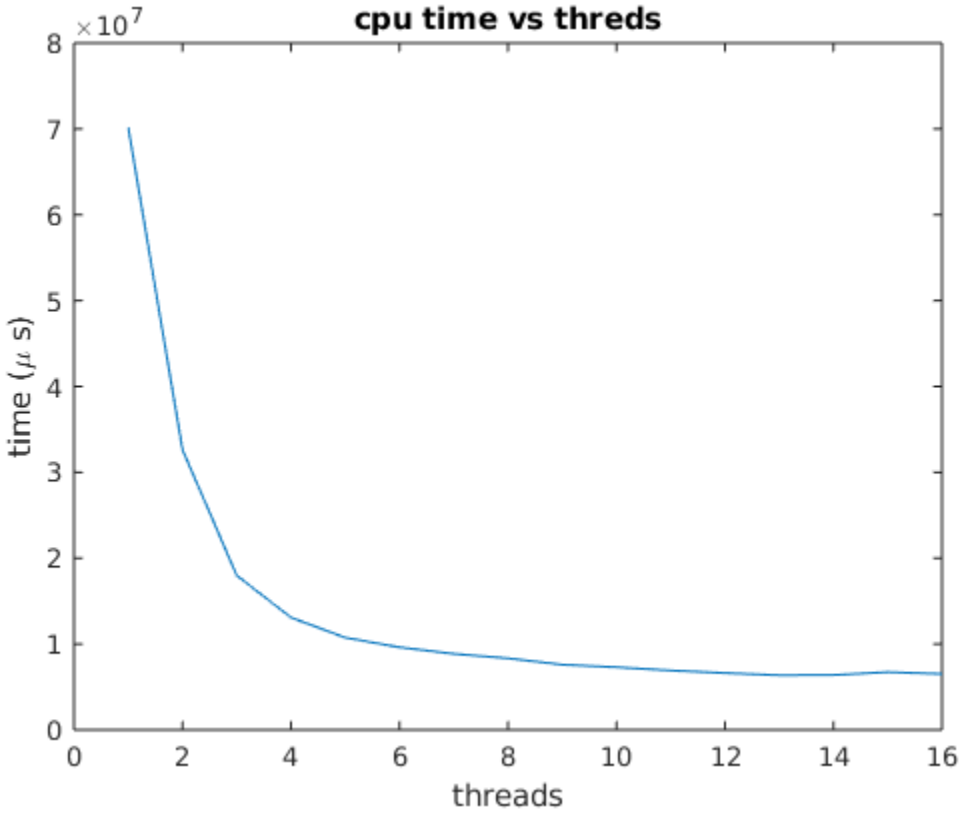
Homework 5 APPM 5720 HPC Spencer Walker

For this assignment I made the main.cpp file print to a file instead of the terminal. I printed the energy at various different times and the total cpu time on two different machines the first was a crappy AMD machine with 16 cores and the second was 32 cores (of many more) on a better Intel machine.

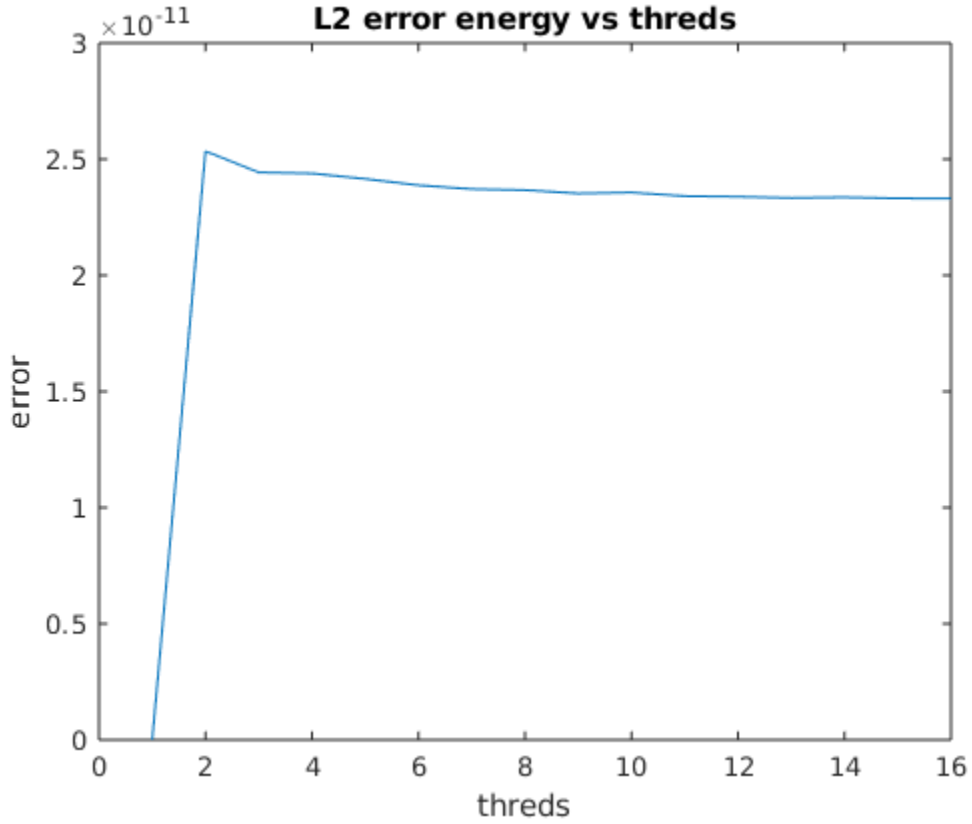
To paralelize the code I inserted a “#pragma omp parallel” for before every loop that could be parallelized in an easy way. For some reason when I wrote the energy loop in parallel the error jumped up from less than 1e-16 to 1e-11. I’m not sure if this is round off error or something worse.

I ran the code with the default boundary conditions and initial/grid parameters that the file was with in the original repository.

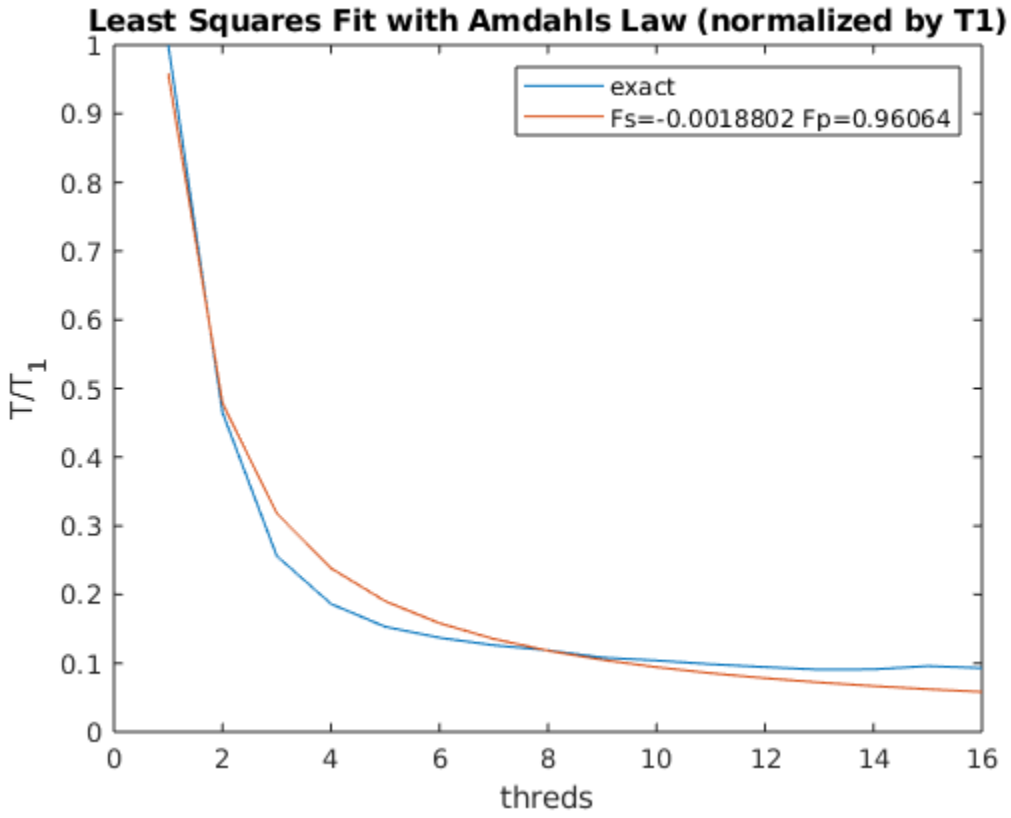
For the AMD machine here is the time vs threads.



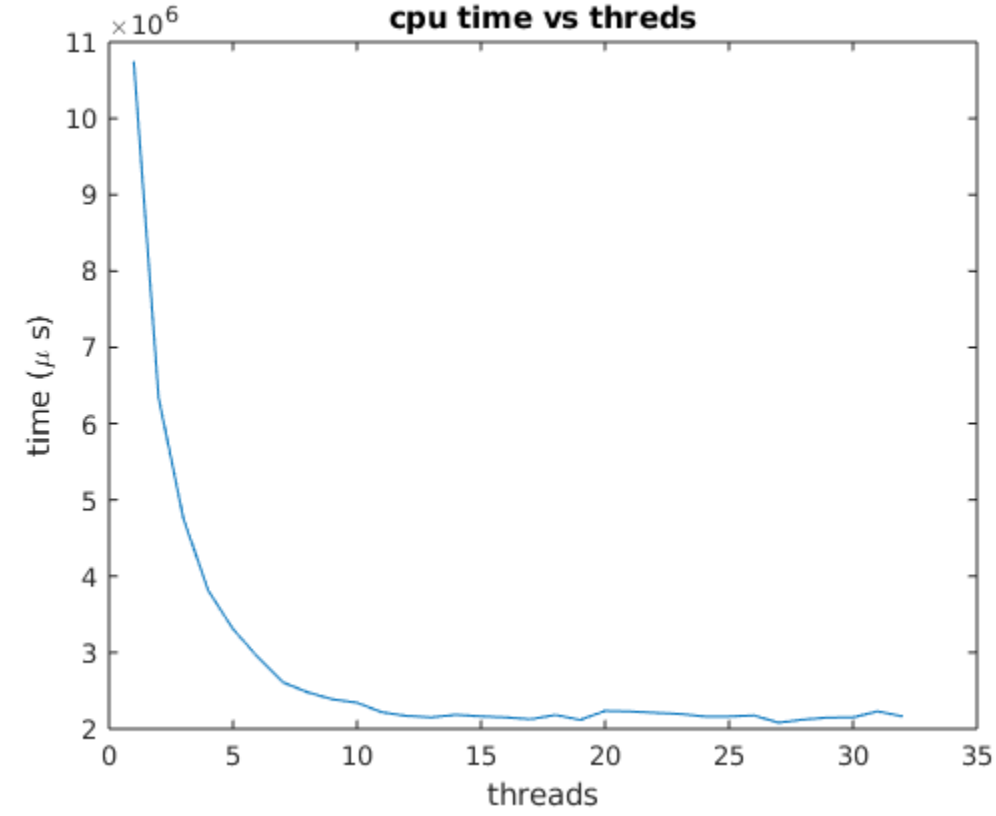
Here's the l2 error in the energy across simulations (using p=1) as reference.

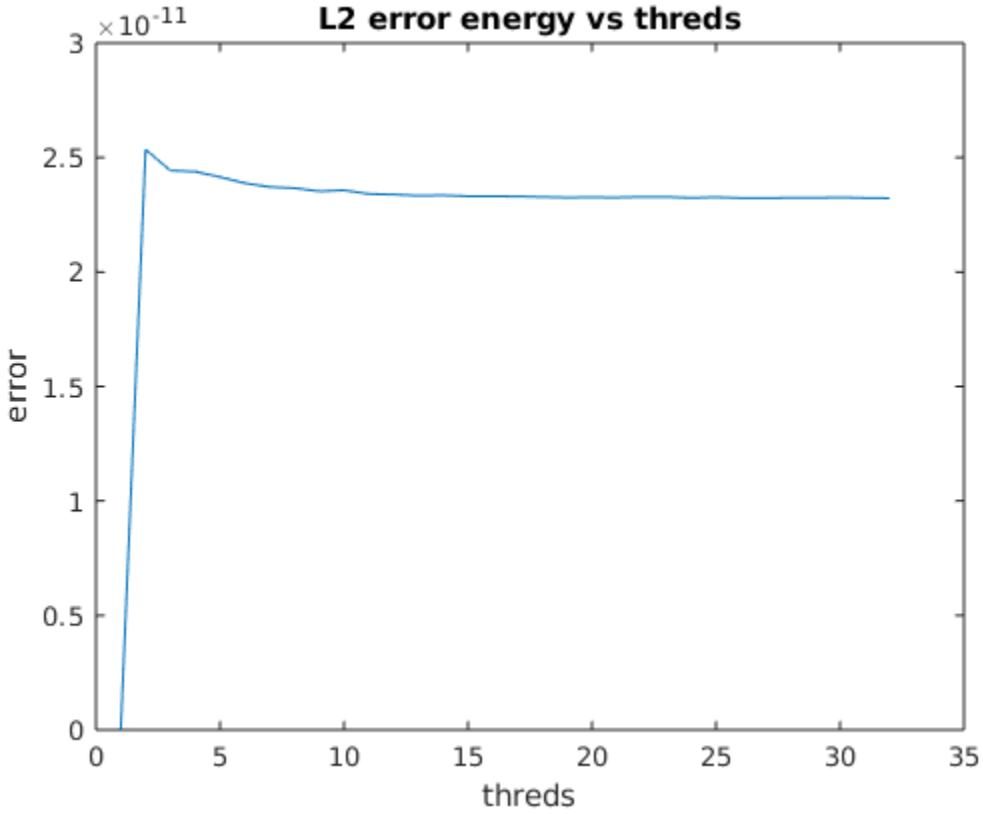


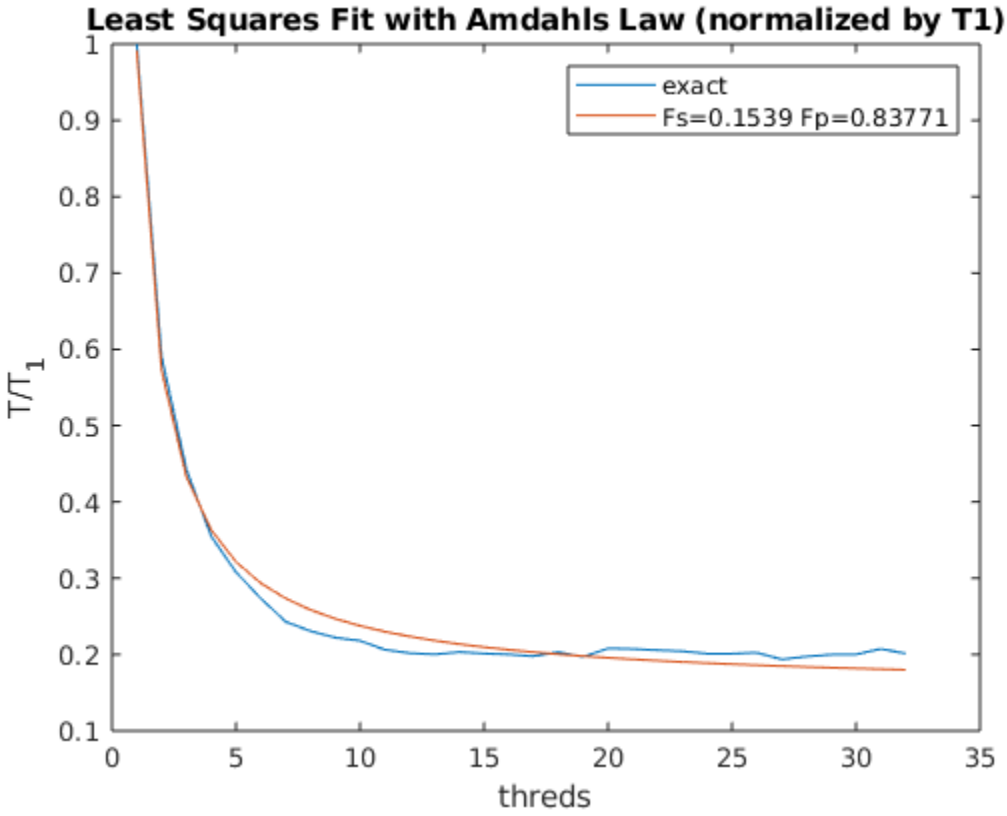
Here's the least squares fit of the data (normalized to T1)



Here's the same thing for the better Intel machine.







An interesting feature is that Fs is larger for the better machine than the AMD machine. I think that this is due to the fact that the Intel cores are each individually faster than the AMD cores so the parallel portions of the code run extremely fast while serial operations like reading/writing files are quite slow. I think that this is probably the bottleneck for these calculations.

As noted above only the trivial loops were parallelized. I believe that many sequential portions can be improved. The simplest speedup would come from writing fewer files and not calculating the energy as often. For example if only the final solution is needed we are wasting a significant time writing files.